Learning for Logical Agents

Outline

- Learning agents
- Inductive learning
- Decision tree learning
- Measuring learning performance

Learning

- Learning is essential for unknown environments
- Useful as a system construction method
- Learning modifies the agent's decision mechanisms

Learning Agents

- Performance element = agent behavior
- Learning element improves performance
- Critic provides feedback
- Problem generator suggests exploration

Learning Element

Design depends on:

- Component to be learned (labels, parameters, moves, actions, ...)
- Representation of that component (Boolean function, linear model, non-linear model, ...)
- Type of feedback available (none, immediate, delayed, uncertain,...)

Types:

- Supervised learning: correct answers given
- Unsupervised learning: no notion of correct answers
- Reinforcement learning: learning to maximise future rewards

Learning as Induction

- Example: Learn a function f from examples
- Example: $(x, f(x))$
- Find ("induce") a hypothesis $h \approx f$
- Training set provides examples

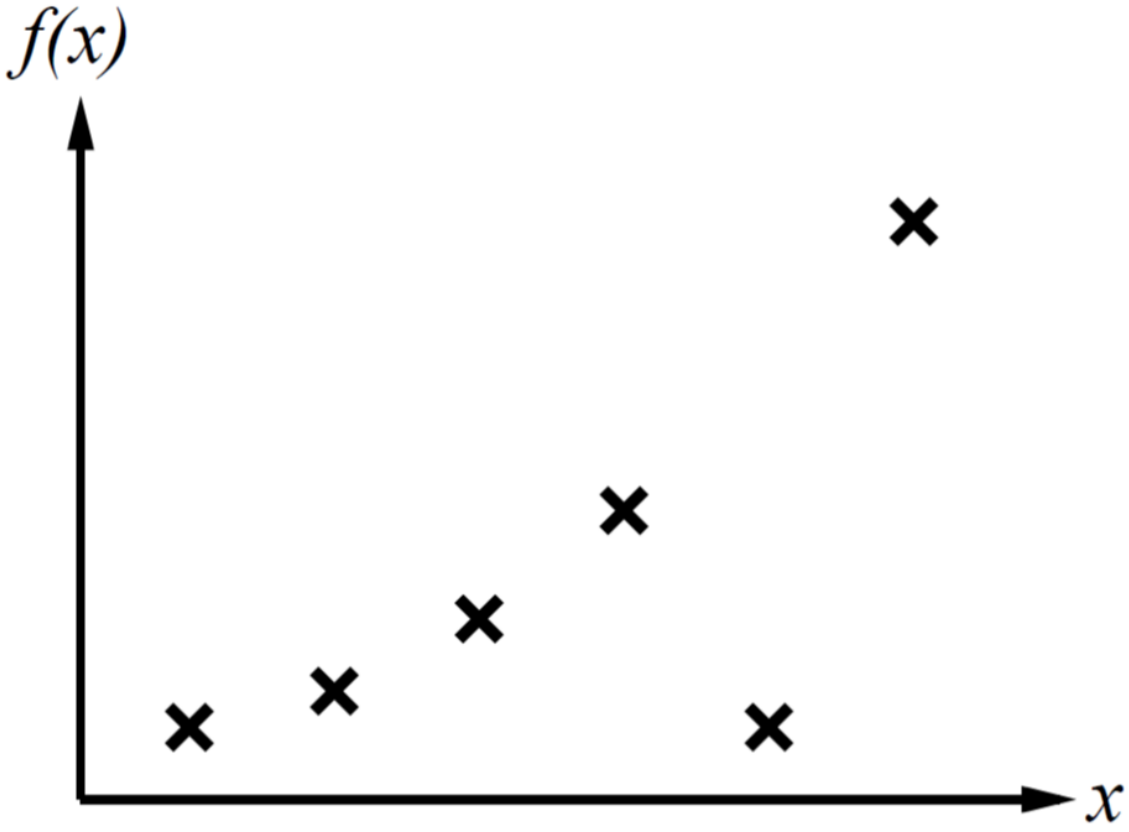
Simplifications:

- No prior knowledge
- Deterministic environment
- Examples given

Inductive Learning Method

- Adjust h to match f on training data
- Consistent hypothesis: agrees on all examples

Key idea: curve fitting



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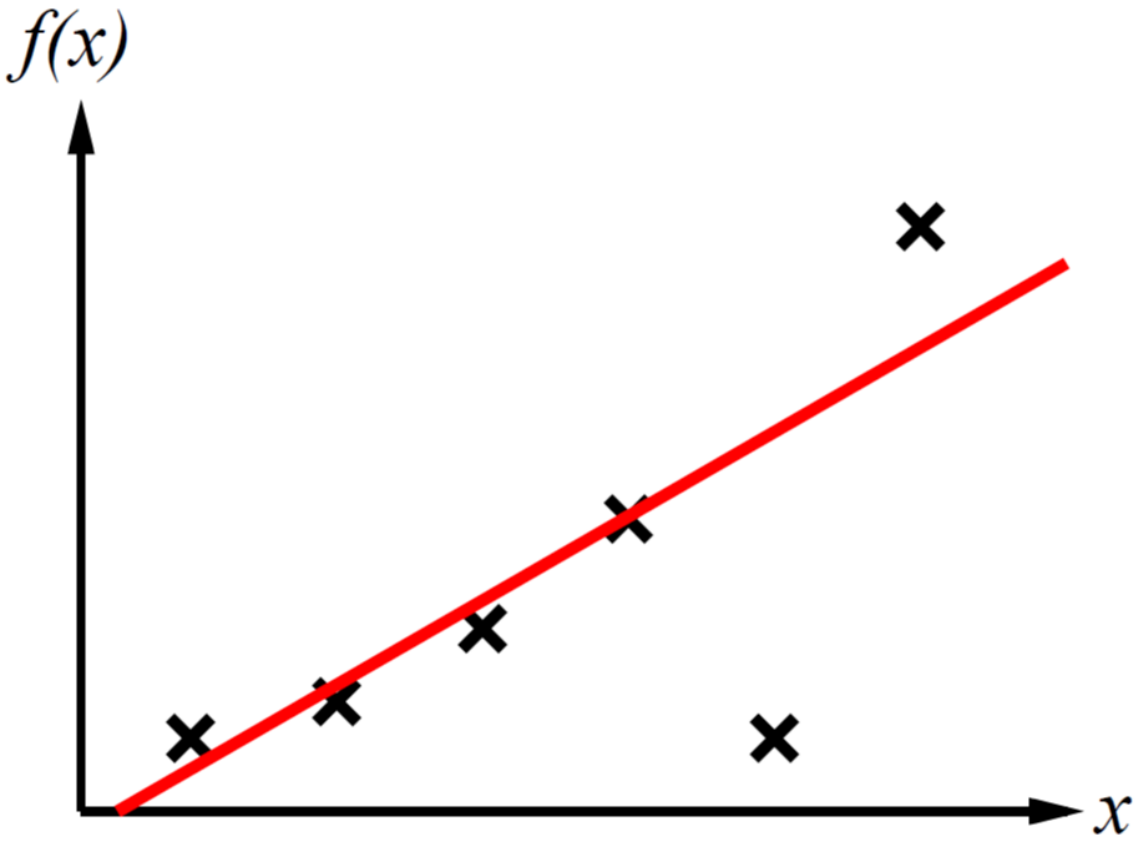

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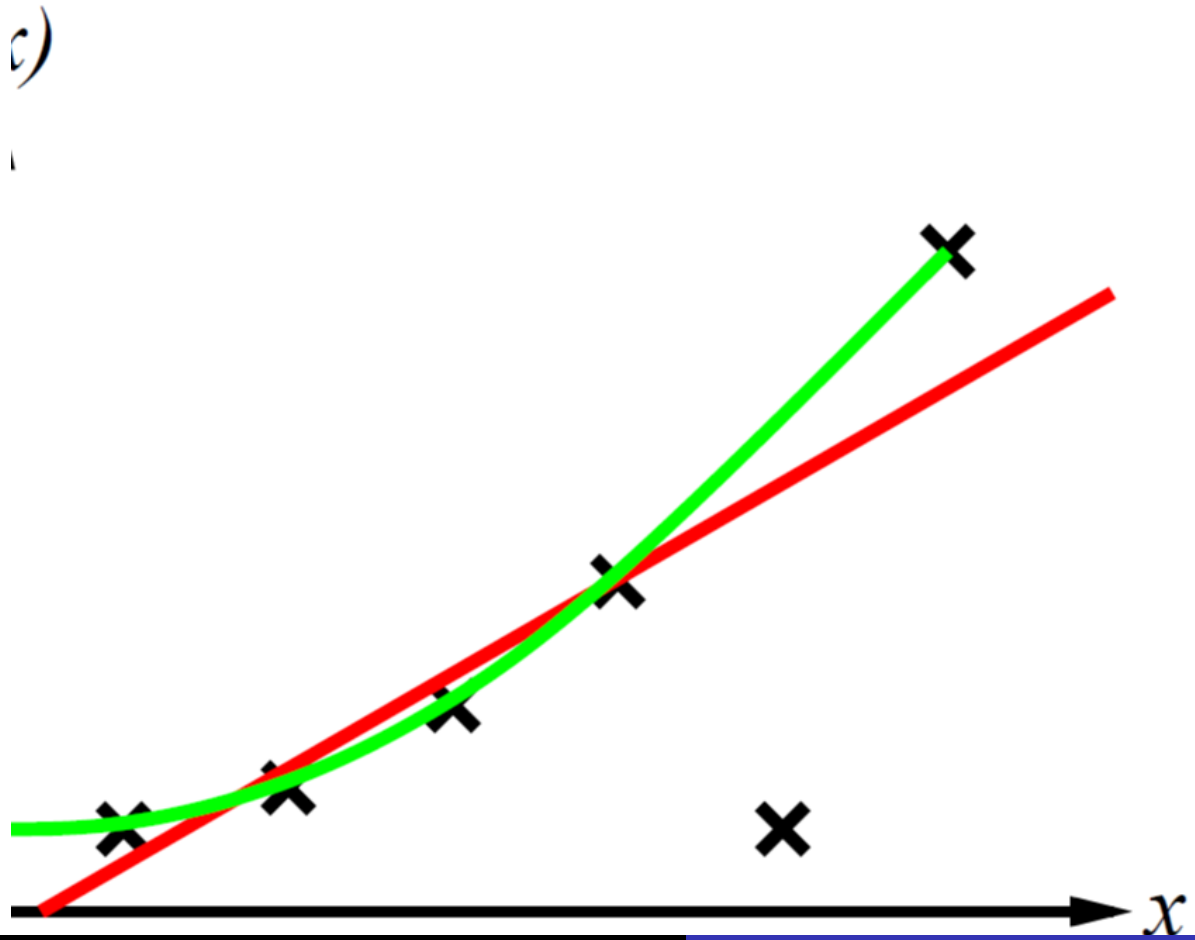

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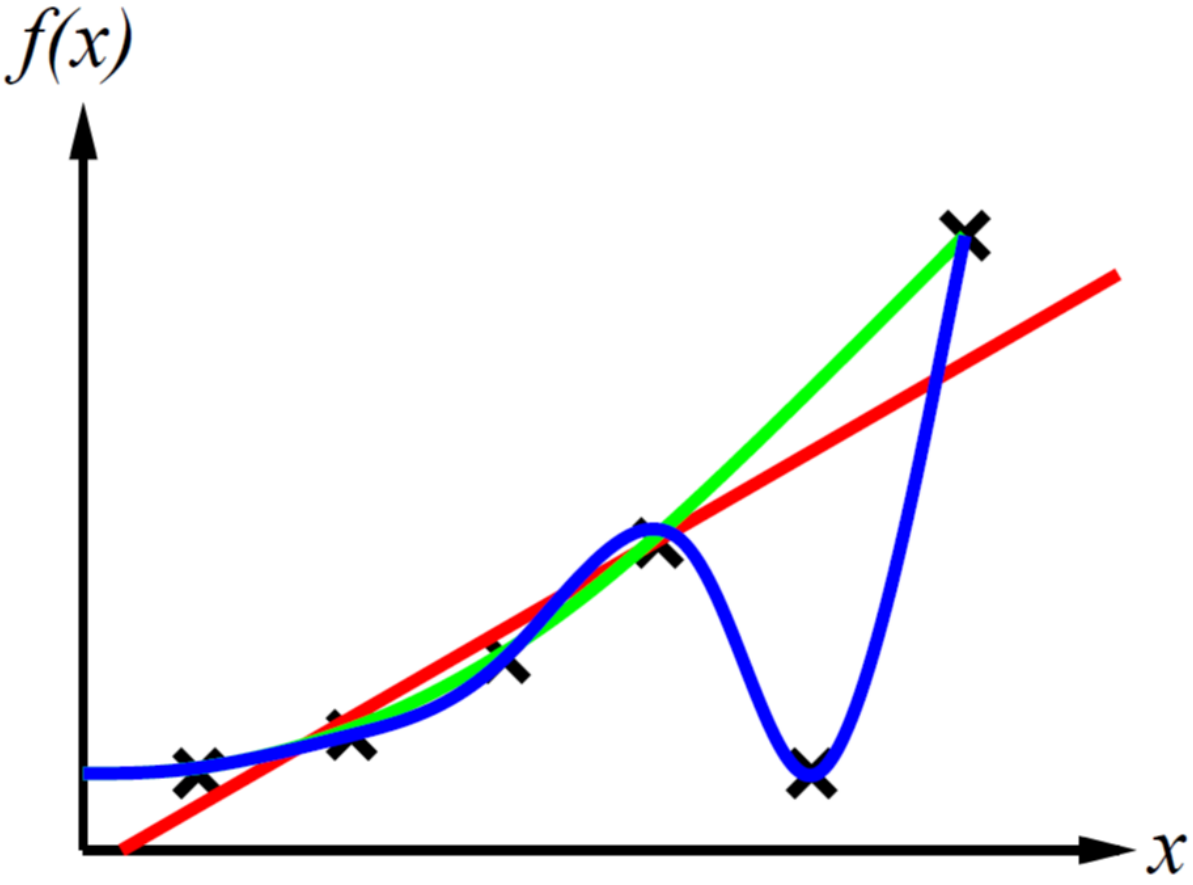


Ockham's Razor

Principle

Prefer hypotheses that are:

- Consistent with data
- As simple as possible

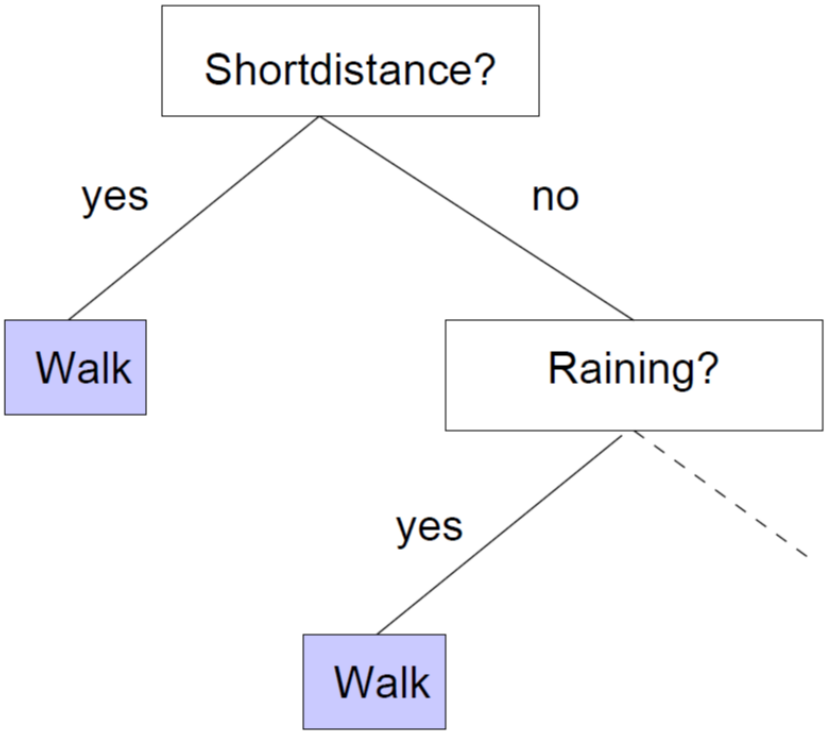
Why is this a good strategy?

Attribute-Based Representation

- Examples described by attributes
- Attributes can be:
  - Boolean
  - Discrete
  - Continuous
- Classification:
  - Boolean valued: Postitive (True) or Negative (False)
  - Multi-valued

Decision Trees

- Representation of hypotheses
- Internal nodes = attributes
- Leaves = classification



Expressiveness

- Decision trees can represent any Boolean function
- Always possible to fit training data
- But may not generalize well

Goal: find compact trees

Hypothesis Space

Number of Boolean functions:

$$2^{2^n}$$

Example:

- $n = 6 \Rightarrow 2^{64}$ functions

Conjunctive hypotheses:

$$3^n$$

- Each hypothesis (or model) can be thought of as having 2 parts: a structure, and parameters
- Hypothesis consisting of linear models:
  - Structure: linear functions
  - Parameters: weights
- Hypothesis consisting of decision trees:
  - Structure: ...
  - Parameters: ...
- A learning agent may need to learn structure, or parameters, or both

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Decision Tree Structure Learning

Goal:

- Find a small tree consistent with data

Method:

- Choose best attribute
- Split data recursively

Choosing Attributes

- Good split $\rightarrow$ pure subsets
- Aim: separate positives and negatives

Idea: maximize information gain

Information (Entropy)

$$H(P_1, \ldots, P_n) = \sum_i -P_i \log_2 P_i$$

- Measures uncertainty
- Higher entropy $\rightarrow$ more uncertainty

Structure Search using Information Gain

$$Entropy(S) = -(p_{pos})\log_2(p_{pos}) - (p_{neg})\log_2(p_{neg})$$

where: p_{pos} os the fraction of instances which are *pos* and p_{neg} is the fraction of instances which are *neg*

- For example:

$$\begin{aligned} Entropy([15+, 4-]) &= -(15/19)\log_2(15/19) - (4/19)\log_2(4/19) \\ &= 0.742 \end{aligned}$$

- Each internal node partitions a set of instances at that node, So, the gain of using an attribute can be defined as the *reduction* in entropy. Choose attribute with the highest gain

$$Gain(S, A) = Entropy(S) - \sum_{v \in Values(A)} \frac{|S_v|}{|S|} Entropy(S_v)$$

What stops an entroy-based search for tree-structure?

Performance Measurement

Holdout performance:

- Performance on "test"-set with same distribution as training data

Learning curve:

- Accuracy vs training size

ROC curve:

- True Positive vs False Postive rate

Summary

- Learning improves performance in unknown environments
- Inductive learning finds hypotheses from data
- Decision trees: use when agent needs to learn Boolean logic representations

  Some instances of the function $\rightarrow$ Learn "good" function
- Evaluate correctness on new data
- Agents can learn more complex representations. For example $Move(x, y)$ if $Piece(x) \wedge Square(y) \wedge Empty(y) \wedge Safe(y)$, $Safe(x)$ if ...
- Homework: How can you modify the simple decision tree learner to:
  - Include probabilities for class labels
  - Predict a number (rather than a class label)